

1 Basic things

Unit impulse | Delta

$$\delta[n] = \begin{cases} 1 & n = 0 \\ 0 & n \neq 0 \end{cases}, \quad \delta_m[n] = \delta[n - m]$$

Unit step

$$u[n] = \begin{cases} 1 & n \geq 0 \\ 0 & n < 0 \end{cases}$$

Exponential decay

$$x[n] = a^n u[n], \quad 0 < a < 1$$

Sinusoidal oscillation

$$x[n] = A \cos(\omega n + \phi), \quad A \in \mathbb{R}, \omega \in \mathbb{R}, \phi \in \mathbb{R}$$

Complex exponential

$$x[n] = A e^{j(\omega n + \phi)}, \quad A \in \mathbb{R}, \omega \in \mathbb{R}, \phi \in \mathbb{R}$$

Energy and Power

- $E_x = \|x\|^2$ (sum or integral of $|x|^2$)
- $P_x = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$
- $E_x < \infty \Rightarrow$ Energy signal ($x \in \ell_2$ or L_2)
- $P_x < \infty$ and $E_x = \infty \Rightarrow$ Power signal

Time shift by $k \in \mathbb{Z}$

$$y[n] = S^k x[n] = x[n + k]$$

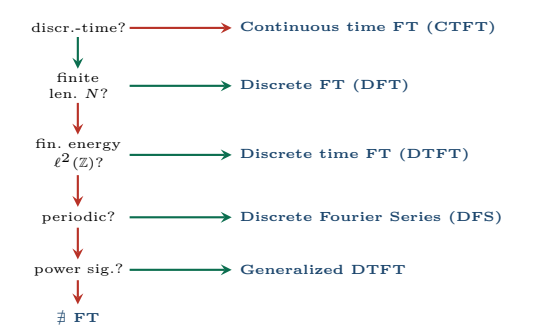
Time reversal

$$y[n] = \mathcal{R}x[n] = S^{-n} x[0] = x[-n]$$

Sum of a sequence

$$\sum_{k=m}^n z^k = \begin{cases} n - m + 1 & z = 1 \\ \frac{z^m - z^{n+1}}{1 - z} & z \neq 1 \end{cases}$$

2 Signal analysis



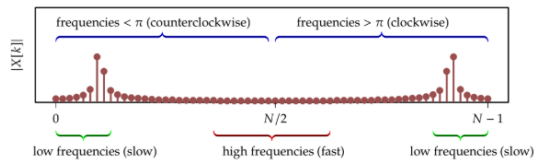
3 DFT: $X[k]$

Fourier basis vectors

$$w_k[n] = e^{j \frac{2\pi}{N} kn}, \quad n, k = 0, \dots, N - 1.$$

Note: $\mathbf{w}_k = \mathbf{w}_{N-k}^*$.

Orthogonality (not orthonormality)

$$\langle \mathbf{w}_h, \mathbf{w}_k \rangle = N \delta[h - k], \text{ where}$$
$$\langle \mathbf{x}, \mathbf{y} \rangle = \sum_{n=0}^{N-1} x^*[n] y[n].$$


DFT matrix form

$$\mathbf{X} = \mathbf{W}\mathbf{x}, \quad \mathbf{x} = \frac{1}{N} \mathbf{W}^H \mathbf{X}, \text{ where } W_N = e^{-j \frac{2\pi}{N}}:$$
$$\mathbf{W} = \begin{bmatrix} W_N^0 & W_N^0 & \cdots & W_N^0 \\ W_N^0 & W_N^1 & \cdots & W_N^{N-1} \\ \vdots & \vdots & \ddots & \vdots \\ W_N^0 & W_N^{N-1} & \cdots & W_N^{(N-1)^2} \end{bmatrix}$$

$\mathbf{W}^H \mathbf{W} = N\mathbf{I}.$

Powers modulo N

$$W_N^n = W_N^{n \bmod N} \text{ (aliasing property).}$$

Small DFT matrices

$$\mathbf{W}_2 = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$
$$\mathbf{W}_4 = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix}$$

4 Properties

A signal $\mathbf{x}$ is **symmetric** if $x[n] = x[N - n]$.
A signal is **Hermitian-symmetric** if $x[n] = x^*[N - n]$.

Symmetry summary

Time domain $x[n]$	Freq. domain $X[k]$
Real	$X[-k] = X^*[k]$
Real & even	Real & even
Real & odd	Imaginary & odd
Complex & even	Complex & even
Complex & odd	Complex & odd

5 DFS: $X[k]$

DFS (Discrete Fourier Series)

For an N -periodic signal $\tilde{x}$:

$$\tilde{\mathbf{X}} = \text{DFS}\{\tilde{\mathbf{x}}\} = \text{DFT}\{\tilde{x}[0], \dots, \tilde{x}[N - 1]\}$$

DFS $\equiv$ DFT: same formulas, different interpretation (explicit periodicity).

Circular shift

$$\tilde{x}[n - n_0] \xleftrightarrow{\text{DFS}} e^{-j \frac{2\pi}{N} n_0 k} \tilde{X}[k]$$

6 DTFT: $X(e^{j\omega})$

As a limit of the DFT

$$X[k] \longrightarrow X(e^{j\omega}) \quad \text{with} \quad \omega = \frac{2\pi k}{N}$$

Symmetry summary

Time domain $x[n]$	Freq. domain $X(\omega)$
Real	$X(-\omega) = X^*(\omega)$
Real & even	Real & even
Real & odd	Imaginary & odd
Complex & even	Complex & even
Complex & odd	Complex & odd

Generalized DTFT (Power Signals)

Via functionals for infinite-energy signals:

Constant & complex exponential

$$\text{DTFT}\{1\} = \delta(\omega) \quad (\text{Dirac delta at } \omega = 0)$$
$$e^{j\omega_0 n} \xleftrightarrow{\text{DTFT}} \delta(\omega - \omega_0)$$

Sinusoids

$$\cos(\omega_0 n) \xleftrightarrow{\text{DTFT}} \frac{1}{2} [\delta(\omega - \omega_0) + \delta(\omega + \omega_0)]$$
$$\sin(\omega_0 n) \xleftrightarrow{\text{DTFT}} \frac{-j}{2} [\delta(\omega - \omega_0) - \delta(\omega + \omega_0)]$$

7 FFT

Complexity comparison

- Naive DFT: $O(N^2)$ multiplications
- FFT: $O(N \log N)$ operations
- For $N = 10,000$: $\sim 40,000$ vs. > 100 million ops

Decimation in time (Cooley-Tukey)

Split $N = 2^L$ input into even/odd: $x_e[n] = x[2n]$, $x_o[n] = x[2n + 1]$, $n = 0, \dots, \frac{N}{2} - 1$
Output halves: $X_a[k] = X[k]$ and $X_b[k] = X[k + \frac{N}{2}]$, $k = 0, \dots, \frac{N}{2} - 1$
Butterfly equations:

$$X_a[k] = X_e[k] + e^{j \frac{2\pi}{N} k} X_o[k]$$
$$X_b[k] = X_e[k] - e^{j \frac{2\pi}{N} k} X_o[k]$$

where X_e , X_o are the $(N/2)$ -point DFTs of x_e , x_o .

Cost breakdown

- $2(N/2)^2 = N^2/2$ mults for $\mathbf{X}_e$ and $\mathbf{X}_o$
- $N/2$ mults by twiddle factors for $\mathbf{X}_o$
- Second half is free (just sign change)

Recursion gives $O(N \log_2 N)$ total.

8 Spectrograms (STFT)

Short-Time Fourier Transform

Analyse a signal $\mathbf{x}$ of length L time-locally:

- Slide a window of size W to get $\lfloor L/W \rfloor$ non-overlapping bins
- Compute a W -point DFT in each bin
- Result: 2D time $\times$ frequency representation

Kronecker delta in spectrogram

DFT of a length- W Kronecker delta $\delta[n - m]$ (even shifted): $|X[k]| = 1$ for all k .
 $\Rightarrow$ In a spectrogram with window W : the bin containing m has **constant magnitude** W across all frequencies; all other bins are **0**.

Window size trade-off

Large W : good frequency resolution, poor time resolution.
Small W : good time resolution, poor frequency resolution.
This is an instance of the time–frequency concentration duality.

9 Filtering I - Digital Filters

LTI Systems

Linearity & time-invariance

Let $y[n] = \mathcal{H}\{x[n]\}$.
Linearity ($\forall \alpha, \beta \in \mathbb{C}, \forall x_1, x_2$):
$$\mathcal{H}\{\alpha x_1[n] + \beta x_2[n]\} \stackrel{?}{=} \alpha \mathcal{H}\{x_1[n]\} + \beta \mathcal{H}\{x_2[n]\}$$

Time-Invariance ($\forall n_0 \in \mathbb{Z}, \forall x$):
$$\underbrace{\mathcal{H}\{x[n - n_0]\}}_{x[\cdot] \leftarrow x[\cdot - n_0]} \stackrel{?}{=} \underbrace{y[n - n_0]}_{n \leftarrow n - n_0}$$

Key consequence: an LTI system cannot create new frequency content.

Impulse Response & Convolution

Impulse response

$\mathbf{h} = \mathcal{H}\delta$. An LTI system is *fully* described by $\mathbf{h}$.

Convolution $\mathbf{y} = \mathbf{x} * \mathbf{h}$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k] h[n - k]$$

Convolution properties

Commutativity: $\mathbf{x} * \mathbf{h} = \mathbf{h} * \mathbf{x}$
Linearity: $\mathbf{x} * (\alpha \mathbf{y} + \beta \mathbf{w}) = \alpha(\mathbf{x} * \mathbf{y}) + \beta(\mathbf{x} * \mathbf{w})$
Time-shift: $\mathbf{x} * (\mathcal{S}^k \mathbf{y}) = \mathcal{S}^k(\mathbf{x} * \mathbf{y})$
Associativity: $(\mathbf{x} * \mathbf{h}) * \mathbf{w} = \mathbf{x} * (\mathbf{h} * \mathbf{w})$
Cascade: $\mathbf{h} = \mathbf{h}_1 * \mathbf{h}_2$; order is irrelevant.

Filter Properties (from h)

FIR vs IIR

- FIR** (Finite Impulse Response): $\mathbf{h}$ has finite support. No feedback; *always* BIBO stable.
- IIR** (Infinite Impulse Response): $\mathbf{h}$ has infinite support. Realizable IIRs use feedback and have a one-sided $\mathbf{h}$.

Causality

- Up to a delay:** $h[n] = 0$ for $0 > n_0 > n$.
- Anticausal:** $h[n] = 0$ for $n > 0$; offline use only.
- Ideal:** h has infinite two-sided support.

BIBO Stability

LTI is BIBO stable $\iff \mathbf{h}$ is absolutely summable:

$$\sum_{k=-\infty}^{\infty} |h[k]| < \infty$$

Constant-Coefficient Difference Equation (CCDE)

$$y[n] = \underbrace{\sum_{k=0}^{M-1} b_k x[n - k]}_{\text{feedforward (FIR part)}} - \underbrace{\sum_{k=1}^{N-1} a_k y[n - k]}_{\text{feedback (IIR part)}}$$

FIR: all $a_k = 0$ for $k \geq 1$.
IIR: at least one $a_k \neq 0$ for $k \geq 1$ — recursion introduces a feedback loop.

Frequency Response

C'est un masque chez les fréquences (ex: lowpass)

$$H(\omega) = \text{DTFT}\{\mathbf{h}[n]\}$$
$$\mathcal{H}\{e^{j\omega_0 n}\} = H(\omega_0) \cdot e^{j\omega_0 n}$$

Convolution theorem

$$y = x * h \iff Y(\omega) = X(\omega) \cdot H(\omega)$$

Filter types by magnitude response

- Lowpass:** passes low freqs ($\omega \approx 0$)
- Highpass:** passes high freqs
- Bandpass:** passes freqs near $\pm\omega_c$
- Allpass:** $|H(\omega)| = \text{const}$ — modifies phase only

Example of filters

Moving Average (MA)

$$y[n] = \frac{1}{M} \sum_{k=0}^{M-1} x[n - k]$$

$h[n] = \frac{1}{M}$ for $0 \leq n < M$, else 0. Causal FIR $\Rightarrow$ always BIBO stable. Big cost.

Leaky Integrator (LI)

$$y[n] = \lambda y[n - 1] + (1 - \lambda) x[n], \quad |\lambda| < 1$$
$$h[n] = (1 - \lambda) \lambda^n u[n] \quad (\text{causal, exponential decay})$$

Causal IIR. Constant cost. Approximates MA of length M with $\lambda = \frac{M-1}{M}$.

Ideal Filters

sinc & rect

$$\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}, \quad \text{sinc}(0) = 1.$$
$$\text{rect}(x) = \begin{cases} 1 & |x| \leq \frac{1}{2} \\ 0 & |x| > \frac{1}{2} \end{cases}$$

DTFT pair: $\frac{\omega_p}{\pi} \text{sinc}\left(\frac{\omega_p}{\pi} n\right) \xleftrightarrow{\text{DTFT}} \text{rect}\left(\frac{\omega}{2\omega_p}\right)$

10 Filtering II - Filter Design

Modulation theorem

$$z[n] = x[n]y[n] \iff Z(\omega) = X(\omega) * Y(\omega)$$

Z-Transform

$$X(z) = \sum_{n=-\infty}^{\infty} x[n] z^{-n}, \quad z \in \mathbb{C}$$

A formal power series in z^{-1} ; existence requires the sum to converge absolutely for a given z .

Fundamental properties

Linearity: $\mathcal{Z}\{\alpha \mathbf{x} + \beta \mathbf{y}\} = \alpha X(z) + \beta Y(z)$
Time-shift: $\mathcal{Z}\{x[n - N]\} = z^{-N} X(z)$

Link to DTFT

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} = \text{DTFT}\{\mathbf{x}\}$$

The DTFT is the z-transform restricted to $|z| = 1$

Transfer Function

Transfer function $H(z)$

From $y[n] = h[n] * x[n] \Rightarrow Y(z) = H(z)X(z)$, so $H(z) = \mathcal{Z}\{\mathbf{h}\} = \frac{Y(z)}{X(z)}$.

Applying $\mathcal{Z}\{\cdot\}$ to the CCDE (time-shift: $x[n - k] \rightarrow z^{-k} X(z)$):

$$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}} = \frac{B(z)}{A(z)}$$

$B(z)$: feedforward (FIR part).
 $A(z)$: feedback (IIR part). FIR: $A(z) = 1$.

Zeros & poles — factored form

$$H(z) = b_0 \frac{\prod_{k=0}^{M-1} (1 - z_k z^{-1})}{\prod_{k=0}^{N-1} (1 - p_k z^{-1})}$$

Zeros z_k : roots of $B(z)$, where $|H| = 0$.
Poles p_k : roots of $A(z)$, where $|H| \rightarrow \infty$.

Stability & ROC

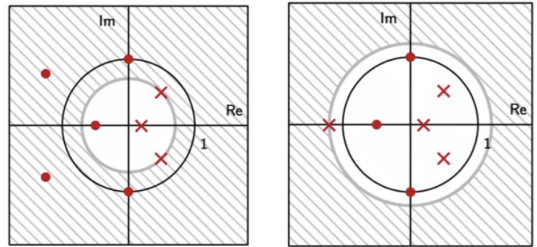
ROC & causality

The ROC is the set of $z \in \mathbb{C}$: $\sum_n |x[n]z^{-n}| < \infty$. Poles bound the ROC — zeros do not affect it.

Causal IIR: ROC = $\{|z| > R\}$: $R = \max_k |p_k|$.
FIR: ROC = $\mathbb{C}$ (except possibly $z = 0$).

BIBO stability

Stable $\iff$ unit circle $|z| = 1$ lies inside the ROC
 $\iff$ all poles strictly inside the unit circle ($|p_k| < 1$).



Stable (unit circle in ROC) Unstable (unit circle not in ROC)

Running Examples in Z-Domain

Moving Average — z-domain

$$H(z) = \frac{1}{N} \cdot \frac{1 - z^{-N}}{1 - z^{-1}} = \frac{1}{N} \prod_{k=1}^{N-1} (1 - e^{j\frac{2\pi}{N}k} z^{-1})$$

$N-1$ zeros at N -th roots of unity (all except $z = 1$); no poles. ROC: all $\mathbb{C}$ except $z = 0$. Always stable (FIR).

Leaky Integrator — z-domain

$$H(z) = \frac{1 - \lambda}{1 - \lambda z^{-1}}, \quad \text{pole at } z = \lambda$$

ROC: $|z| > \lambda$ (causal). Stable $\Leftrightarrow |\lambda| < 1$ (pole inside unit circle). As $\lambda \rightarrow 1$: pole approaches unit circle, bandwidth narrows.

11 Filtering III - Stochastic and Adaptive SP

Rappel de proba +

- Auto-correlation** $R_X = E[x^2]$
- Cross-correlation** $R_{XY} = E[xy]$
- Covariance** $C_{XY} = E[(x - m_x)(y - m_y)]$
- if zero-mean** $C_{XY} = R_{XY}$
- Uncorrelated** $E_{XY} = E[x]E[y]$
- Independent** $f_{XY}(x, y) = f_X(x)f_Y(y)$

Deterministic vs Stochastic

Deterministic: $x[n]$ fully described by a formula.
Stochastic: a machine of ∞ RVs indexed by n ; each run produces one **realization** $\check{x}[n]$.
 k -th order description:

- **1st:** $f_{X[n]}(x[n])$
- **2nd:** $f_{X[n]X[m]}(x[n], x[m])$

Full description (all orders) is impossible.

Stationarity

1 - Strict: all joint PDFs are time-shift invariant.
2 - WSS (Wide-Sense stationarity):

1. $E[x[n]] = m_x$
2. $E[x[n]x[m]] = r_x[n - m]$

Properties: $r_x[-k] = r_x[k]$; $r_x[0] = E[x[n]^2] = P_x \geq 0$.

12 A/D I - Interpolation & Sampling

The Continuous-Time World

CT vs. DT paradigm

D: $x_d[n] \in \ell_2(\mathbb{Z})$, $n \in \mathbb{Z}$, $\omega \in [-\pi, \pi]$, DTFT
A: $x_c(t) \in L_2(\mathbb{R})$, $t \in \mathbb{R}$ [s], $f \in \mathbb{R}$ [Hz], CTFT

Use cases

- $x(t) \rightarrow$ [processing] $\rightarrow y(t)$
- $x[n] \rightarrow$ [processing for synthesis] $\rightarrow y(t)$
- $x(t) \rightarrow$ [processing for analysis] $\rightarrow y[n]$

Real-world frequency

$f = \# \text{rep/s} = \text{Hz}$; $\Omega = 2\pi f$

CTFT (more info in cs)

- NOT PERIODIC
- NO LIMIT FREQUENCY

BandLimited (BL) functions

A CT signal is F_S -BL $\Leftrightarrow X(f)$ for $|f| > F_S/2$

Support of a function

The interval where the function is non-trivially zero

Sampling theorem (Shannon–Nyquist)

Any signal $x(t)$ whose highest frequency component is F_N Hz can be sampled with **no loss of information** using a sampling frequency:

$$F_s \geq 2F_N \quad (\text{i.e. } T_s \leq \frac{1}{2F_N})$$

Interpolation $x[n] \rightarrow x(t)$

Interpolation requirement

- decide on T_s (sampling period)
- make sure $x(nT_s) = x[n]$
- make sure $x(t)$ is smooth

Interpolations pros and cons

Polynomial: $+ C^\infty$ (max. smooth) – kernels $L_n^{(N)}$ depend on N

Local: $+ \text{kernel indep. of } N$ – lacks smoothness

Remarkable result

$$\lim_{N \rightarrow \infty} L_n^{(N)}(t) = \text{sinc}(t - n)$$

Sinc interpolation

$$x(t) = \sum_{n \in \mathbb{Z}} x[n] \text{sinc}(t - n)$$

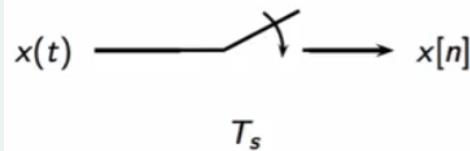
$\text{sinc}(m - n) = \delta[m - n]$, $\forall m, n \in \mathbb{Z}$
CTFT pair ($T_s = 1/F_s$):

$$\varphi(t) = \text{sinc}\left(\frac{t}{T_s}\right) \longleftrightarrow \Phi(f) = \frac{1}{F_s} \text{rect}\left(\frac{f}{F_s}\right)$$

13 A/D II - Aliasing

Raw Sampling

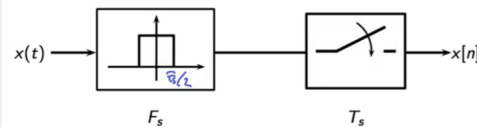
$$x[n] = x(nT_s)$$



Sinc Sampling

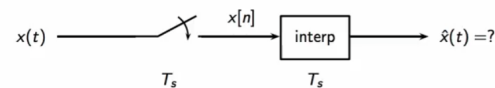
$$x[n] = \left\langle \text{sinc}\left(\frac{t - nT_s}{T_s}\right), x(t) \right\rangle$$

$$x[n] = (\text{sinc}_{T_s} * x)(nT_s)$$



Aliasing

Sinc Sampling



Map the frequency into $[-F_s/2 ; F_s/2]$ to make the signal F_s -BL.

14 A/D III - Multirate

Downsampling by N "DTFT pov"

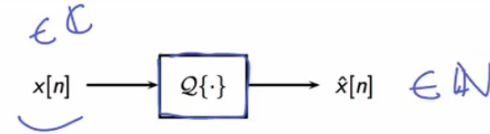
Copier à tout $2\pi/N$. Multiplier les indices par N .

Upsampling by N "DTFT pov"

Montrer la périodicité. Diviser les indices par N .

15 A/D IV - A/D and D/A conversion

Quantization schemes



16 Quick Reference

Geometric sum

$$\sum_{n=0}^{N-1} r^n = \frac{1 - r^N}{1 - r}, \quad r \neq 1$$

Used in rectangular DFT derivation; also the orthogonality of Fourier vectors (set $r = e^{j\frac{2\pi}{N}(k-h)}$, $k \neq h$).

Euler & trig

- $(e^{j\theta})^* = e^{-j\theta}$
- $\cos(a + b) = \cos a \cos b - \sin a \sin b$
- $\sin(a + b) = \sin a \cos b + \cos a \sin b$
- $\sin a \sin b = \frac{1}{2} [\cos(a - b) - \cos(a + b)]$
- $\cos a \cos b = \frac{1}{2} [\cos(a - b) + \cos(a + b)]$
- $\sin a \cos b = \frac{1}{2} [\sin(a + b) + \sin(a - b)]$

Complex number tricks

- $z_1 = z_2^* \Rightarrow |z_1| = |z_2|$ ($|z| = |z^*|$ always)
- $z + z^* = 2 \operatorname{Re}\{z\}$, $z - z^* = 2j \operatorname{Im}\{z\}$
- $z = z^* \iff z \in \mathbb{R} \iff \operatorname{Im}\{z\} = 0$
- $|z_1 z_2| = |z_1| |z_2|$, $\angle(z_1 z_2) = \angle z_1 + \angle z_2$
- $|z|^2 = z \cdot z^*$

Inner product in $\mathbb{C}^N$

Coefficients: $X[k] = \langle \mathbf{w}_k, \mathbf{x} \rangle$ (projection onto basis).

DTFT inner product in $L_2([-\pi, \pi])$

$$\langle \mathbf{F}, \mathbf{G} \rangle = \frac{1}{2\pi} \int_{-\pi}^{\pi} F^*(\omega) G(\omega) d\omega$$

Set $\{e^{-j\omega n}\}_{n \in \mathbb{Z}}$ forms an orthogonal basis.

$$x[n] = \langle e^{j\omega n}, X(\omega) \rangle$$

Key warnings

DFT normalization

Analysis: no $1/N$ factor. **Synthesis:** $1/N$ factor.
Convention: $\mathbf{W}^H \mathbf{W} = N \mathbf{I}$.

Phase wrapping

Phase plots show $\angle X[k] \in [-\pi, \pi]$ (wrapped). Apparent discontinuities are $\pm\pi$ jumps, not real.

Real signal DFT

For real $\mathbf{x}$: $X[k] = X^*[N - k]$. Only first $\lfloor N/2 \rfloor + 1$ coefficients are independent. Magnitude symmetric around $k = N/2$.

Circular vs. linear shift

DFT/DFS time shifts are always **circular** (modulo N). For aperiodic finite signals, multiplying by phase in freq domain gives circular shift.